

Proceedings

Lateral-Force AFM Study of Graphene Thickness Effects for Reduced Steel Friction

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Lateral-force atomic force microscopy (LF-AFM) is a key characterization method in friction studies of graphene, a promising nanomaterial for low-friction coatings[1]. LF-AFM enables nanoscale friction measurements, essential for probing friction properties of small areas of the coating. Two key factors for graphene's performance as a friction reducing layer are graphene grain size[2] and thickness[3]. Prior friction studies of graphene on steel, featuring transferred graphene, found that thicker, large grain size graphene has the lowest friction. The study presented here uncovers the effects of thickness and grain size for graphene that is directly grown on 8620 steel by chemical vapor deposition (CVD). This CVD coating is a mixture of multilayer graphene (MLG), and few-layer graphene (FLG). MLG areas have large grain sizes and a thicker coating, while FLG areas have smaller grains, and a thinner coating. The CVD coating thickness can be controlled to be dominantly MLG or FLG by changing CVD growth parameters[4]. The results of the nanoscale characterization of MLG and FLG areas in this study suggest that CVD grown MLG on low-alloy steel is superior to FLG. This is because the MLG areas are thicker with larger grain sizes, supporting MLG dominant, low-friction CVD graphene coatings for steel. Raman and AFM data were obtained with a Horiba HR800 and a Nanosurf FlexAFM in lateral force mode respectively.

Raman spectroscopy and mapping were used first to identify MLG, FLG, and uncoated areas for friction characterization using LF-AFM. A typical spectrum for MLG is shown in Fig. 1(a) in blue, and a typical spectrum for FLG is shown in teal. The CVD conditions resulted in a MLG dominant coating, with some FLG areas and a small amount of single-layer graphene which was mixed with the FLG. This is displayed by the mapped 30 x 30 μm area of the coating in Fig. 1(b). The map is 85% MLG (blue pixels) and 12% FLG areas (teal and green pixels) with pixel colors based on the graphene thickness determined by 2D:G peak ratio. The Raman map was used to select spots for friction characterization by pixel color.

Fig. 1(c) is a topographic AFM map of the same area mapped in (b). The steel surface after CVD exhibits height differences that are less than one micron (average roughness ~ 80 nm), and the MLG mainly covers the tops and sides of the largest, highest features. This is beneficial, because MLG effectively acts as a lubricating barrier for contact points during wear. FLG covers lower topographic areas and areas with finer textures, while gaps in the graphene coating are most often at the lowest, flattest topographic points.

The LF-AFM micrograph in Fig. 2(a) presents a 1 x 1 μm spot at the edge of an uncoated and FLG coated area, highlighting the contrast of the graphene coating with the uncoated steel. The FLG has much lower friction than the uncoated steel, evidenced by the dark contrast in the friction force micrograph. However, the friction is higher at graphene grain boundaries (brighter contrast). The FLG grains of varying thickness also show different friction forces, with thicker areas having lower friction. Fig. 2(b) illustrates that the overall friction for a MLG dominant area is much lower than FLG, because the uniform, thicker graphene is a more effective barrier when compared to FLG. Additionally, the larger grain size minimizes the effects of grain boundaries which raise friction. The MLG therefore acts as a better lubricating layer, by more effectively covering the steel in a thicker graphene layer, without areas of higher friction caused by grain boundaries or thin graphene. Statistical analysis of the coatings was performed by selecting 30 spots (10 each), 500 x 500 nm in size, from MLG, FLG and uncoated areas. These were compared to mirror polished 8620 without CVD graphene (7 spots). The calculated friction forces summarized in Fig. 2(c), indicate that MLG reduced friction the most, by 71% compared to polished steel.

This study demonstrates that CVD graphene coatings are able to reduce friction for low-alloy steels, where the large grain size and thicker graphene of MLG play critical roles for friction reduction.

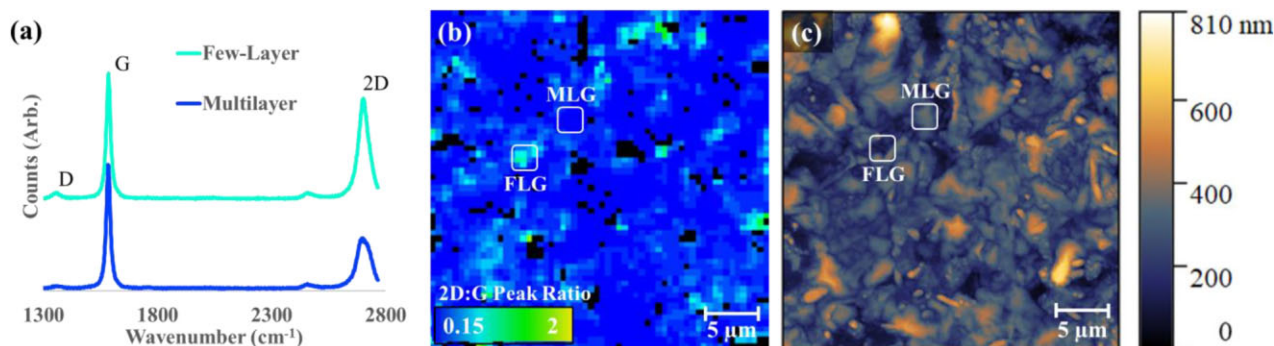


Fig 1. (a) Raman spectra from the MLG dominant (blue) graphene coating with some FLG (teal) areas. (b) Raman heatmap showing high coverage of MLG

(blue pixels) compared to FLG areas (teal and green), and uncoated areas (black). The MLG is thicker with larger grain sizes, and correspond to the raised features of the surface in (c), acting as a lubricating barrier. (c) AFM topography map of the same area from (b) showing the textured MLG areas.

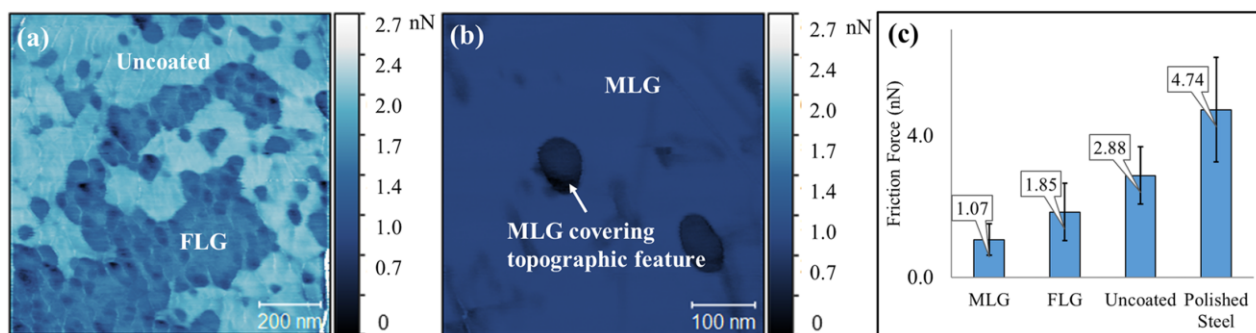


Fig 2. (a) LF-AFM micrograph showing that FLG has much lower friction than uncoated steel. The small grain sizes result in higher friction at grain boundaries, and differences in graphene thickness affect friction for individual grains. (b) LF-AFM micrograph of MLG which is uniform, and has lower friction compared to FLG. The larger grain sizes minimize the effects of grain boundaries, and the thick graphene has lower overall friction. (c) Average friction force of MLG and FLG compared to uncoated areas and as-polished steel. MLG coatings have an excellent 76% reduction of friction on average compared to mirror-polished steel. Note, due to the micron-scale roughness of MLG areas, a 500 x 500 nm scan size was used to minimize topographic artifacts.

References

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